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MOVABLE DISPLAY DEVICE

Abstract

A movable display device includes: a support; a display member; and a driver configured to rotationally move the display member to be placed into a displaying position and a non-displaying position. A restrictor configured to restrict a movement of the display member upon bringing the display member into at least one of the displaying position or the non-displaying position is provided. The restrictor includes a roller provided on or with either of the support, or the display member or an attached member which rotationally moves with the display member; a restriction surface provided on or with another of the support, or the display member or the attached member; and a spring configured to press the roller against the restriction surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on and claims priority to Japanese Patent Application No. 2024-036156 filed on Mar. 8, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Field of the Invention

[0002] The present disclosure relates to a movable display device capable of rotating a display member to set the display member to be in a displaying position or a non-displaying position, and restricting the display member not to move while in at least one of the displaying position or the non-displaying position.

2. Description of the Related Art

[0003] Japanese Laid-Open Patent Application No. 2014-8825 discloses a monitor storage device. This monitor storage device opens and closes a monitor stored in a storage section provided on the ceiling of a passenger aircraft cabin. The monitor is supported in the storage section to be able to rotate about a rotation axis. A motor is provided in the monitor storage device. By the power of the motor, the monitor is operated to switch between a closed state, in which the monitor is parallelly stored in the storage section with respect to the storage section, and an opened state, in which the monitor is usable by being opened at an angle of about 110 degrees with respect to the storage section. A toggle spring is provided between a rotation section that rotates together with the monitor and the storage section, and the monitor is biased to be in the closed state as well as to be in the opened state by an elastic force of the toggle spring.

[0004] Japanese Laid-Open Patent Application No. 2019-21696 discloses a panel angle switching structure. This panel angle switching structure is used as a vehicle navigation device, and includes a body, a panel storage section fixed to the front of the body, and a panel stored in the panel storage section. A slide base that moves in a front-rear direction by an electric drive device is provided inside the body. When the slide base moves backward, the panel is put into an upright state, and when the slide base moves forward, the panel is put into an inclined state. A wall section is provided to the front of the slide base, and a shaft provided on the panel is rotatably supported by the wall section, and a plate spring fixed to the shaft contacts the wall section. In this structure, since the panel is held by a frictional force acting between the wall section and the plate spring, generation of mechanical undesirable noise due to vibration can be prevented.

SUMMARY

[0005] A movable display device is disclosed including a support, a display member, and a driver configured to rotationally move the display member to be placed into a displaying position and a non-displaying position. A restrictor configured to restrict a movement of the display member upon bringing the display member into at least one of the displaying position or the non-displaying position is provided. The restrictor includes a roller provided on or with either of the support, or the display member or an attached member which rotationally moves with the display member, a restriction surface provided on or with another of the support, or the display member or the attached member, and a spring configured to press the roller against the restriction surface.

[0006] The movable display device is disclosed, wherein the spring is a cantilever that is elastically deformable, and the roller is supported on a free end of the cantilever. In conjunction with a moving direction of the roller relative to the restriction surface upon reaching at least one of the displaying position or the non-displaying position by the display member, the moving direction upon the reaching being a restriction proceeding direction, preferably the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further

ahead in the restriction proceeding direction than a fixed end of the cantilever, while the roller is pressed against the restriction surface.

[0007] The movable display device of the present disclosure includes a displaying-side restrictor configured to restrict a movement of the display member upon reaching the displaying position by the display member, and a non-displaying-side restrictor configured to restrict the movement of the display member upon reaching the non-displaying position by the display member. Upon restricting the display member by the displaying-side restrictor or by the non-displaying-side restrictor, the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further ahead in the restriction proceeding direction than the fixed end of the cantilever.

[0008] For example, in the movable display device of the present disclosure, the driver includes: a slide member in which the display member is rotatably supported; a power source configured to reciprocate the slide member to move the slide member between a predetermined displaying position and a predetermined non-displaying position; and a display-position switcher configured to rotationally move the display member toward the displaying position by using a moving force of the slide member upon moving the slide member toward the predetermined displaying position, and to rotationally move the display member toward the predetermined non-displaying position by using the moving force of the slide member upon moving the slide member toward the non-displaying position, wherein the displaying-side restrictor and the non-displaying-side restrictor are arranged with an interval in a moving direction of the slide member.

[0009] In the movable display device of the present disclosure, the cantilever and the roller are supported by the display member or an attached member that rotationally moves with the display member, and for both cases where the display member is restricted by the displaying-side restrictor and by the non-displaying-side restrictor, by switching a direction of the cantilever upon moving of the slide member between the predetermined displaying position and the predetermined non-displaying position, the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further ahead in the restriction proceeding direction than the fixed end of the cantilever.

[0010] A movable display device is disclosed, which is provided with a support; a slide member supported to reciprocate in a direction crossing a gravitational direction with respect to the support; [0011] a display member rotatably supported by the slide member; a power source configured to reciprocate the slide member toward a predetermined displaying position and a predetermined non-displaying position; a display-position switcher configured to rotationally move the display member toward the displaying position, in which an end of the display member opposite to a rotation support side of the display member faces the gravitational direction, by using a moving force of the slide member upon moving the slide member toward the predetermined displaying position, and to rotationally move the display member toward the predetermined non-displaying position, in which the end of the display member is rotationally lifted upward, by using the moving force of the slide member upon moving the slide member toward the non-displaying position; a displaying-side restrictor configured to restrict a movement of the display member in the displaying position; and a non-displaying-side restrictor configured to restrict the movement of the display member in the non-displaying position, wherein upon moving the slide member between the predetermined displaying position and the predetermined non-displaying position, the movement of the display member is not restricted either by the displaying-side restrictor or the non-displaying-side restrictor.

[0012] The movable display device of the present disclosure is provided with a guide cam configured to guide the display-position switcher upon moving the slide member, wherein the guide cam is provided with an inclined guide portion which is inclined with respect to a moving direction of the slide member, and the display-position switcher is guided by the inclined guide portion upon moving the slide member between the predetermined displaying position and the

predetermined non-displaying position, and upon guiding the display-position switcher by the inclined guide portion, the display member is not restricted either by the displaying-side restrictor or the non-displaying-side restrictor.

[0013] In the movable display device of the present disclosure, the displaying-side restrictor and the non-displaying-side restrictor preferably include a roller provided on or with either of the support, or the display member or an attached member which rotationally moves with the display member; a restriction surface provided on or with another of the support, or the display member or the attached member; and a spring which presses the roller against the restriction surface.

[0014] The movable display device of the present disclosure may be configured to include a holder configured to hold the display member rotationally moved to be placed into the non-displaying position, so as not to move in the gravitational direction by using the moving force of the slide member upon moving the slide member toward the predetermined non-displaying position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a perspective view illustrating an overall structure of the movable display device according to an embodiment of the present disclosure, in which a display member is in a displaying position;

[0016] FIG. 2 is a perspective view illustrating the overall structure of the movable display device according to the embodiment of the present disclosure, in which the display member is in a non-displaying position;

[0017] FIG. 3 is a partially enlarged view of FIG. 1 and is a partial perspective view mainly illustrating a structure of a support and a driver;

[0018] FIG. 4 is an exploded perspective view illustrating the driver as illustrated in FIG. 3;

[0019] FIG. 5 is a side sectional view illustrating a state in which the display member is in a displaying position, in a movable display device according to the embodiment of the present disclosure;

[0020] FIG. 6 is a side sectional view illustrating an operation state in which the display member is in a transition from the displaying position to the non-displaying position, in a movable display device according to the embodiment of the present disclosure;

[0021] FIG. 7 is a side sectional view illustrating a state in which the display member is in the non-displaying position, in the movable display device according to the embodiment of the present disclosure;

[0022] FIG. 8A is a partially enlarged side sectional view illustrating an operation of a displaying-side restrictor;

[0023] FIG. 8B is another partially enlarged side sectional view illustrating the operation of the displaying-side restrictor;

[0024] FIG. 9A is a partially enlarged side sectional view illustrating an operation of a non-displaying-side restrictor;

[0025] FIG. 9B is another partially enlarged side sectional view illustrating the operation of the non-displaying-side restrictor;

[0026] FIG. 10 is a partial perspective view illustrating a holder for holding the display member and an operation in the middle of holding an end of the display member;

[0027] FIG. 11 is a partial perspective view illustrating the holder for holding the display member, in a state in which the holding of the end of the display member is completed;

[0028] FIG. 12A is an explanatory diagram for illustrating a restricting operation of the non-displaying-side restrictor; and

[0029] FIG. 12B is another explanatory diagram for illustrating the restricting operation of the

displaying-side restrictor.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0030] In the monitor storage device described in Japanese Laid-Open Patent Application No. 2014-8825, since a large rotational force from the toggle spring acts on the monitor when the monitor is brought into the opened state and the closed state, an impact force and an impact sound are likely to be generated when the monitor switches to each state. In addition, a large driving force is required to invert the toggle spring when the monitor is rotationally moved from the opened state to the closed state and when the monitor is rotationally moved from the closed state to the opened state. Therefore, a load always acts on a driver for rotating the monitor, and abrasion or the like of the driver is likely to occur.

[0031] In the panel angle switching structure described in Japanese Laid-Open Patent Application No. 2019-21696, the frictional force between the wall portion and the plate spring always acts as a load when the panel is switched from the upright state to the inclined state, such that the loss of power is large and abrasion is likely to occur between the wall portion and the plate spring.

[0032] The present disclosure provides a movable display device capable of stabilizing the display member in at least one of a displaying position or a non-displaying position, and capable of suppressing an increase in the load for stabilizing the display member in each position.

[0033] A movable display device **1** according to an embodiment of the present disclosure is installed on the ceiling in a vehicle cabin. Alternatively, the movable display device **1** can be installed on the ceiling in a cabin of a train, on the ceiling in a cabin of an aircraft, on the ceiling in a building or the like.

[0034] In each drawing, a Y1-Y2 direction is a front-rear direction, a Y1 direction is a rear direction and a Y2 direction is a front direction. The Y1 direction is toward a predetermined displaying position, and the Y2 direction is toward a predetermined non-displaying position. An X1-X2 direction is a transverse direction, an X1 direction is a left direction, and an X2 direction is a right direction. A Z1-Z2 direction is a vertical direction, a Z1 direction is an upward direction, and a Z2 direction is a downward direction. When the movable display device **1** is installed on the ceiling in the vehicle cabin, the Z2 direction is a gravitational direction.

[0035] As illustrated in FIGS. **1** and **2**, the movable display device **1** includes supports **2a** and **2b**. The supports **2a** and **2b** are support frames that extend linearly parallel to each other in the front-rear direction (Y1-Y2 direction). The support **2a** is located on a left side (X1 side), and the support **2b** is located on a right side (X2 side). When the movable display device **1** is fixed to the ceiling in the vehicle cabin, the supports **2a** and **2b** are fixed in parallel to an X-Y plane, which is substantially a horizontal plane.

[0036] The movable display device **1** includes a display member **3**. The display member **3** includes a rectangular casing **4**, and the casing **4** includes a front surface **4a** and a back surface **4b**. A display panel such as a liquid crystal display panel and an electroluminescent panel is cased in the casing **4**. The display screen of the display panel appears on the front surface **4a** of the casing **4**. The casing **4** includes a rotation support **4c** and an end portion **4d** opposite to the rotation support side where the rotation support **4c** is located. In FIGS. **1** and **5**, the display member **3** is in the displaying position, and the front surface **4a** of the casing **4** is facing forward (Y2 direction) and slightly downward, and the end portion **4d** is facing downward (Z2 direction) in the gravitational direction. In FIGS. **2** and **7**, the display member **3** is in the non-displaying position (storage position), the end portion **4d** of the casing **4** is lifted upward (Z1 direction), and the front surface **4a** and the back surface **4b** are substantially parallel to the horizontal plane (X-Y plane).

[0037] As illustrated in FIGS. **1** and **2**, a left driver **10a** is provided between the support **2a** and the display member **3**, and a right driver **10b** is provided between the support **2b** and the display member **3**. The drivers **10a** and **10b** are the structures and mechanisms of the members described in FIGS. **1** and **2**, except for the supports **2a** and **2b**, the display member **3**, and the displaying-side restrictor **40a** and the non-displaying-side restrictor **40b**, which will be described in the following.

The left driver **10a** and the right driver **10b** are plane-symmetrical in a left-right direction (X1-X2 direction). The right driver **10b** is provided with a motor as a drive source.

[0038] FIGS. **3** and **4** illustrate details of the left driver **10a**. The left driver **10a** is provided with an outer bracket **11** and an inner bracket **12**.

[0039] The outer bracket **11** is bent at a right angle to form a vertical support portion **11a** and a horizontal support portion **11b**. As illustrated in FIGS. **1** and **2**, the vertical support portion **11a** is fixed to an inner surface of the support **2a** facing the right side (X2 side) by screwing. A fixed rail **13** is fixed to a surface facing the right side of the vertical support portion **11a** by screwing. The fixed rail **13** extends linearly in the front-rear direction (Y1-Y2 direction). The left driver **10a** is provided with a slide member **14**. As illustrated in FIG. **4**, the slide member **14** includes an outer slide portion **15** and an inner slide portion **16**. In the front of the slide member **14**, the outer slide portion **15** and the inner slide portion **16** are fixed to each other by a front connecting shaft **17**, and in the rear, the outer slide portion **15** and the inner slide portion **16** are fixed to each other by a rear connecting shaft **18**. The outer slide portion **15** is a movable rail and is slidably held by the fixed rail **13** in the front-rear direction. The slide member **14** composed of the outer slide portion **15** and the inner slide portion **16**, which are fixed to each other, is supported to reciprocate linearly in the front-rear direction (Y1-Y2 direction) with respect to the support **2a**.

[0040] As illustrated in FIGS. **1** and **2**, the left driver **10a** and the right driver **10b** are connected by a connecting drive shaft **21**. As illustrated in FIG. **3**, the connecting drive shaft **21** is supported inside a long guide slot **12a** formed in the front-rear direction in the inner bracket **12**, and a left (X1 side) end of the connecting drive shaft **21** is rotatably supported by a bearing **24** fixed to the inner slide portion **16**. As illustrated in FIG. **4**, a rack member **23** is fixed to a lower surface of the horizontal support portion **11b** of the outer bracket **11**. A pinion gear **22** is fixed to a left end of the connecting drive shaft **21**, and the pinion gear **22** is engaged with rack gear teeth formed in a lower portion of the rack member **23**.

[0041] As illustrated in FIGS. **1** and **2**, the right driver **10b** is also provided with the slide member **14**, and a right end of the connecting drive shaft **21** is rotatably supported by the bearing **24** fixed to the inner slide portion **16** of the slide member **14**. The pinion gear **22** is also fixed to the right end of the connecting drive shaft **21**, and the pinion gear **22** is engaged with the rack gear teeth of the rack member **23** fixed to the outer bracket **11** of the right driver **10b**. In the right driver **10b**, a motor serving as the drive source and the gears are supported by the inner slide portion **16**, the pinion gear **22** is driven by the motor's power, and thus, the connecting drive shaft **21** is rotatably driven. As a result, in both the left driver **10a** and the right driver **10b**, the left and right sliding members **14** reciprocate in the front-rear direction (Y1-Y2 direction) in synchronization by the rotational force of the pinion gears **22** engaged with the rack members **23**.

[0042] As illustrated in FIGS. **4** and **5**, an attached member **25** is fixed to both the right and left ends of the rotation support **4c** of the casing **4** provided in the display member **3**. A support hole **25a** is opened in each of the left and right attached members **25**, and the front connecting shaft **17** fixed to the slide member **14** is inserted into the support hole **25a**. The support holes **25a** are elongated holes extending so as to draw an arc. When the support holes **25a** are rotated while their movements are being restricted by the front connecting shafts **17**, the display member **3** rotationally moves to switch between the displaying position and the non-displaying position.

[0043] As illustrated in FIGS. **1** and **2**, each of the left driver **10a** and the right driver **10b** is provided with a mechanical linkage **30** serving as a display-position switcher. Each mechanical linkage **30** includes a drive linkage **31** and a driven linkage **32**. The drive linkage **31** and the driven linkage **32** are rotatably connected together by a connection pin **33**. As illustrated in FIGS. **3** and **4**, the drive linkage **31** is positioned inside a space between the outer slide portion **15** and the inner slide portion **16** of the slide member **14** to overlap with a cam member **38**, which will be described in the following. A drive pin **34** projecting into the space is fixed to the inner slide portion **16**, and the drive pin **34** is rotatably inserted into a hole **31a** opening at an end of the drive linkage **31**.

Since the drive linkage **31** rotationally moves when the slide member **14** linearly moves in the front-rear direction, a slight difference is generated between a linear movement path of the drive pin **34** and a movement path of the hole **31a**. In order to solve this difference, an inner diameter of the hole **31a** is formed slightly larger than a diameter of the drive pin **34**. A driven pin **35** is fixed to both sides of the casing **4** of the display member **3**. A hole **32a** is opened at an end of the driven linkage **32** on both sides of the casing **4**, and the driven pin **35** is rotatably inserted into the corresponding hole **32a**.

[0044] As illustrated in FIG. 4, although the cam member **38** is positioned between the outer slide portion **15** and the inner slide portion **16** of the slide member **14**, the cam member **38** is integrally formed with a fixed projection **38a** projecting to the left side (X1 side), and the fixed projection **38a** is fixed to the vertical support portion **11a** of the outer bracket **11**. When the connecting drive shaft **21** and the pinion gear **22** rotate, the left and right slide members **14** move in the front-rear direction, but the cam member **38** does not move because it is fixed to the outer bracket **11**. As illustrated in FIGS. 5 to 7, a guide cam (long guide hole or guide groove) **39** is formed in the cam member **38**, and a position control projection **36** fixed to a rear (Y1 side) end of the drive linkage **31** is inserted into the guide cam **39**. The position control projection **36** is a slide pin that slides in the guide cam **39** or a roller that rolls in the guide cam **39**. A rear end of the guide cam **39** is a displaying-position guide portion **39a** formed in parallel with a moving direction of the slide member **14**, and the front end of the guide cam **39** is a non-displaying-position guide portion **39b** formed in parallel with the moving direction of the slide member **14**. Between the displaying-position guide portion **39a** and the non-displaying-position guide portion **39b** is an inclined guide portion **39c** inclined with respect to the moving direction of the slide member **14**. As illustrated in FIG. 5, a roller **27** and

[0045] a rotation support **28** supporting the roller **27** are supported by the attached member **25** fixed to the rotation support **4c** of the casing **4** forming the display member **3**. As illustrated enlarged in FIGS. 8A to 9B, the rotation support **28** is formed by bending an elastically deformable metal plate, and is integrally formed with a cantilever (cantilever as a plate spring) **28a** as a spring and a holder **28b** formed at a free end of the cantilever **28a**. The roller **27** is a roller and is rotatably supported by a support shaft **29** fixed to the holder **28b**. The attached member **25** is formed of a metal plate and includes a support plate portion **25b**. A fixing block **26** is fixed to an inner surface of the support plate portion **25b**, and a fixed end of the cantilever **28a** is fixed to the fixing block **26** by a screw **26a**. The fixed end of the cantilever **28a** may be directly fixed to the casing **4** of the display member **3** instead of being supported by the attached member **25**.

[0046] As illustrated in FIGS. 1, 2, and 5, the movable display device **1** is provided with a displaying-side restrictor **40a** and a non-displaying-side restrictor **40b**. The displaying-side restrictor **40a** is provided on the left and right sides of the movable display device **1**, and the non-displaying-side restrictor **40b** is also provided on the left and right sides of the movable display device **1**. The displaying-side restrictor **40a** and the non-displaying-side restrictor **40b** are provided with an interval therebetween in the front-rear direction (Y1-Y2 direction). The displaying-side restrictor **40a** is provided on the rear side (Y1 side), and the non-displaying-side restrictor **40b** is provided on the front side (Y2 side). Each of the displaying-side restrictors **40a** provided on the left and right sides of the movable display device **1** includes a part of the outer bracket **11** fixed to the corresponding support **2a** or **2b**, and the non-displaying-side restrictor **40b** includes a part of a restricting plate **43** fixed to a front portion in each of the supports **2a** and **2b**.

[0047] As enlarged and illustrated in FIGS. 8A and 8B, the displaying-side restrictor **40a** includes a displaying-side restriction surface **41** formed on a lower surface of a front end (Y2-side end) of the horizontal support portion **11b** of the outer bracket **11**, the cantilever **28a**, and the roller **27**. The displaying-side restriction surface **41** includes a horizontal restriction surface **41a** that is parallel to the X-Y plane, and an inclined restriction surface **41b** that is continuous with the horizontal restriction surface **41a** and inclined upward toward the front (Y2 direction). As enlarged and

illustrated in FIGS. 9A and 9B, the non-displaying-side restrictor 40b includes a non-displaying-side restriction surface 42 formed on the lower surface of a rear end (Y1-side end) of the restricting plate 43, and the cantilever 28a, and the roller 27. The non-displaying-side restriction surface 42 includes a horizontal restriction surface 42a that is parallel to the X-Y plane, and an inclined restriction surface 42b that is continuous with the horizontal restriction surface 42a and inclined upward toward the rear (Y1 direction).

[0048] FIGS. 10 and 11 illustrate a holder 50 configured to hold the casing 4 of the display member 3 so as not to move in the gravitational direction (Z2 direction) when the display member 3 is rotationally lifted to the non-displaying position. The holder 50 includes a holder projection 51 provided on a support side and a lifting portion 53 provided on the end portion 4d of the casing 4 of the display member 3. As also illustrated in FIG. 5, the holder projection 51 is a holder roller and is rotatably supported by a roller shaft 52 fixed to the vertical support portion 11a of the outer bracket 11. The lifting portion 53 is formed in a part of the casing 4 of the display member 3 and includes a lift-guide surface 53b that is continuous with a lifting surface 53a. The lift-guide surface 53b is inclined gradually away from the back surface 4b of the casing 4 toward the rotation support 4c of the casing 4.

[0049] Next, a movement of the movable display device 1 will be described. In FIGS. 1, 3, and 5, in each of the left driver 10a and the right driver 10b, the slide member 14 is guided by the fixed rail 13 and is moved and stopped at the predetermined displaying position, which is a rear most position (Y1 direction) guided by the fixed rail 13. At this time, the display member 3 is rotationally moved to the displaying position by the operation of the mechanical linkage 30, the casing 4 of the display member 3 is rotationally moved counterclockwise with the front connecting shaft 17 as a fulcrum, and the end portion 4d of the casing 4 is directed in the gravitational direction (Z2 direction). In the movable display device 1, the supports 2a and 2b are fixed to the ceiling of the vehicle cabin or the like, and the display screen appearing on the front surface 4a of the casing 4 is directed forward and obliquely downward such that an occupant in the vehicle cabin can visually view the display screen.

[0050] As illustrated in FIG. 8A, when the slide member 14 stops at the rear predetermined displaying position and the display member 3 rotationally moves counterclockwise to be in the displaying position, in the displaying-side restrictor 40a, the roller 27 contacts the horizontal restriction surface 41a of the displaying-side restriction surface 41 integrally formed with the outer bracket 11. The roller 27 is pressed against the horizontal restriction surface 41a by a flexural elastic force of the cantilever 28a, which is a spring, and by a reactive force of a pressing force, a movement of the display member 3 is restricted so as not to be moved by rattling or the like.

[0051] When an operation for switching the display member 3 from the displaying position to the non-displaying position is performed, a rotational force is applied to the pinion gear 22 fixed to the right end of the connecting drive shaft 21 from the motor mounted on the slide member 14 of the right driver 10b. Since the connecting drive shaft 21 is driven counterclockwise, the pinion gear 22 rotates against the rack teeth of the rack member 23 in both the left driver 10a and the right driver 10b, and the slide member 14 moves forward (in the Y2 direction). The operation of the driver when the slide member 14 moves forward is as illustrated in order from FIGS. 5 to 7. Since the slide member 14 is provided with the drive pin 34 connected to the front connecting shaft 17 supporting the display member 3 and the drive linkage 31, the display member 3 and the drive linkage 31 move forward together with the slide member 14. In this process, the position control projection 36 provided in the drive linkage 31 moves the inclined guide portion 39c of the guide cam 39, and the drive linkage 31 rotates counterclockwise with the drive pin 34 as the fulcrum. This rotational force is transmitted to the casing 4 of the display member 3 via the driven linkage 32, and the display member 3 rotationally moves clockwise with the front connecting shaft 17 as the fulcrum. As illustrated in FIGS. 2 and 7, when the slide member 14 reaches the predetermined non-displaying position, which is a forward end point of the movement, a position sensor (not

illustrated) detects the reach and stops the movement of the motor, the forward movement of the slide member **14** ends, and the display member **3** is put into the non-displaying position (storage position), which is a substantially horizontal position.

[0052] As illustrated in FIG. **9B**, when the slide member **14** stops at the predetermined non-displaying position in the front and the display member **3** rotationally moves clockwise to be in the non-displaying position, in the non-displaying-side restrictor **40b**, the roller **27** comes into contact with the horizontal restriction surface **42a** of the non-displaying-side restriction surface **42** integrally formed with the restricting plate **43**. Since the roller **27** is pressed against the horizontal restriction surface **42a** by the flexural elastic force of the cantilever **28a**, the movement of the display member **3** is restricted by the reactive force of the pressing force so as not to be moved by rattling or the like. When the slide member **14** moves to the predetermined non-displaying position, as illustrated in FIGS. **9A** and **9B**, since the roller **27** rolls from the inclined restriction surface **42b** to the horizontal restriction surface **42a**, the load acting on the slide member **14** from the non-displaying-side restrictor **40b** becomes a rolling resistance, and thus, the movement of the display member **3** can be restricted in the non-displaying position with a relatively small load. This is the same when the slide member **14** moves toward the predetermined displaying position and the roller **27** rolls from the inclined restriction surface **41b** to the horizontal restriction surface **41a**, as illustrated in FIGS. **8A** and **8B**.

[0053] Immediately after the slide member **14** starts moving forward from the predetermined displaying position as illustrated in FIG. **5**, the position control projection **36** provided on the drive linkage **31** is positioned on the displaying-position guide portion **39a** of the guide cam **39**. When the slide member **14** reaches the predetermined non-displaying position as illustrated in FIG. **7**, the position control projection **36** is positioned on the non-displaying-position guide portion **39b** of the guide cam **39**. Since the displaying-position guide portion **39a** and the non-displaying-position guide portion **39b** of the guide cam **39** extend in parallel with the moving direction of the slide member **14**, the load caused by friction or the like when the position control projection **36** moves the displaying-position guide portion **39a** and non-displaying-position guide portion **39b** is small. As illustrated in FIG. **6**, when the slide member **14** moves between the predetermined displaying position and the predetermined non-displaying position, the position control projection **36** moves along the inclined guide portion **39c**, and a forward moving force of the position control projection **36** acts on an inclined portion of the guide cam **39**, such that a moving load acting on the position control projection **36** increases.

[0054] In the displaying-side restrictor **40a**, the roller **27** is pressed against the horizontal restriction surface **41a** when the position control projection **36** is positioned at the displaying-position guide portion **39a** of the guide cam **39**. In the non-displaying-side restrictor **40b**, the roller **27** is pressed against the horizontal restriction surface **42a** when the position control projection **36** is positioned at the non-displaying-position guide portion **39b** of the guide cam **39**. When the position control projection **36** is moving through the inclined guide portion **39c** of the guide cam **39**, the roller **27** is separated from both the horizontal restriction surface **41a** and the horizontal restriction surface **42a**, such that the display member **3** is not subjected to a restricting force from either the displaying-side restrictor **40a** or the non-displaying-side restrictor **40b**. In the movable display device **1**, the load acting on the motor of the display member **3** is extremely large because the end portion **4d** of the display member **3** rotationally moves in the gravitational direction and an antigravitational direction. Therefore, when the position control projection **36** is moving through the inclined guide portion **39c** of the guide cam **39** and the load by the guide cam **39** is large, the restricting force acting on the display member **3** is released from both the displaying-side restrictor **40a** and the non-displaying-side restrictor **40b**, thereby suppressing excessive operation load from being applied to the movable display device **1**.

[0055] When the slide member **14** moves forward (Y2 direction) to reach the predetermined non-displaying position, in the non-displaying-side restrictor **40b**, the roller **27** is guided by the inclined

restriction surface **42b** of the non-displaying-side restriction surface **42** to roll to reach the horizontal restriction surface **42a**, as illustrated in FIG. **9A**. During this time, the roller **27** is pressed against the inclined restriction surface **42b** and the horizontal restriction surface **42a** by the flexural elastic force of the cantilever **28a**. FIG. **12A** is an explanatory diagram of the load when the roller **27** is pressed against the non-displaying-side restriction surface **42**. As illustrated in FIG. **7**, while the slide member **14** moves to the predetermined non-displaying position, the display member **3** and the attached member **25** rotationally move in a clockwise direction. Therefore, before the roller **27** hits the non-displaying-side restriction surface **42**, the cantilever **28a** is rotationally moved clockwise, and the cantilever **28a** is put into a state where the cantilever **28a** extends obliquely toward the front direction (Y2 direction) of the slide member **14**. FIG. **12A** illustrates a center line O of the cantilever **28a** when it is assumed that the cantilever **28a** is not bent. When the slide member **14** and the display member **3** reach the non-displaying position, and the moving direction of the roller **27** relative to the non-displaying-side restriction surface **42** is a restriction proceeding direction Sy2, at a time point before the roller **27** hits the inclined restriction surface **42b** of the non-displaying-side restriction surface **42**, a free end side of the center line O is further inclined in a direction toward the restriction proceeding direction Sy2 than a fixed end side. When the roller **27** is pressed against the inclined restriction surface **42b** and the horizontal restriction surface **42a**, the cantilever **28a** is deflected and deformed toward the restriction proceeding direction Sy2, and a free end **28d** continues to be further inclined toward the restriction proceeding direction Sy2 than a fixed end **28c**.

[0056] In FIG. **12A**, when the moving force of the slide member **14** acting on the axis center of the support shaft **29** of the roller **27** is F_{y2} , and a contact reactive force acting on the axis center of the support shaft **29** from a contact portion T between the roller **27** and the restriction surface is F_g , a movement vector F_r , which is the vector sum of a vector F_{y2} and a vector F_g , acts on the axis center of the support shaft **29**. In addition, the moment M_r of a spring reactive force in which the cantilever **28a**, which is bent toward the restriction proceeding direction Sy2, tries to return elastically, acts in a counterclockwise direction.

[0057] Since the moment of the movement vector F_r is in an opposite direction to the moment M_r of the reactive force, a pressing force F_t acting on the non-displaying-side restriction surface **42** from the roller **27** is reduced at the contact portion T, and a rolling friction resistance f_{y2} of the roller **27** is also reduced. In contrast to FIG. **12A**, when the roller **27** is pressed against the inclined restriction surface **42b** and the horizontal restriction surface **42a**, when it is assumed that the cantilever **28a** is bent to the left direction in the drawing, which is opposite to the restriction proceeding direction Sy2, and that the cantilever **28a** is bent toward the Y1 side of the center line O, the moment M_r of the spring reactive force becomes clockwise, that is the same direction as the moment of the movement vector F_r . Therefore, the pressing force F_t increases, and thus, the rolling friction resistance f_{y2} increases. In the movable display device **1** of the embodiment, the moving load due to an elastic restoring force of the cantilever **28a** can be reduced by keeping the cantilever **28a** inclined toward the front of the restriction proceeding direction Sy2 and bending toward the front of the restriction proceeding direction Sy2.

[0058] When the slide member **14** moves backward (Y1 direction) from the non-displaying position of the display member **3** as illustrated in FIGS. **2** and **7** and returns to the predetermined displaying position as illustrated in FIG. **5**, the display member **3** is rotationally moved counterclockwise. Since the rotation support **28** supported by the attached member **25** also rotates counterclockwise, the cantilever **28a** is positioned obliquely upward toward the rear (Y1 direction). In this position, as illustrated in FIG. **8B**, the roller **27** supported by the cantilever **28a** is guided to the inclined restriction surface **41b** of the displaying-side restriction surface **41**, and as illustrated in FIG. **8A**, the roller **27** rotates while being pressed against the horizontal restriction surface **41a**.

[0059] When the slide member **14** and the display member **3** reach the displaying position as illustrated in FIG. **12B**, a restriction proceeding direction Sy1, which is the relative moving

direction of the roller **27** with respect to the displaying-side restriction surface **41**, becomes the Y1 direction, and the center line O when the cantilever **28a** is in a free state continues to incline such that the free end side is located further toward the restriction proceeding direction Sy1 than the fixed end side. When the roller **27** is pressed against the displaying-side restriction surface **41**, the cantilever **28a** bends toward the restriction proceeding direction Sy1. Also, at this time, the moment due to the movement vector Fr acting on the axis center of the support shaft **29** of the roller **27** and the moment Mr of the reactive force, which is a force trying to restore the bending deformation of the cantilever **28a**, are in opposite directions. Therefore, the pressing force Ft acting on the displaying-side restriction surface **41** from the roller **27** is reduced at the contact portion T, and a rolling friction resistance fy1 of the roller **27** is also reduced.

[0060] FIG. **10** illustrates a state immediately before the slide member **14** moves forward (Y2 direction) to reach the predetermined non-displaying position as illustrated in FIG. **7**. At this time, in the holder **50**, the lift-guide surface **53b** provided close to the end portion **4d** of the back surface **4b** of the casing **4** of the display member **3** contacts an upper portion of the holder projection **51** supported by the outer bracket **11**. When the display member **3** together with the slide member **14** moves forward (Y2 direction), the lift-guide surface **53b** starts to climb up the holder projection **51**, and when the slide member **14** stops at the non-displaying position, the lifting surface **53a** fully climbs up the holder projection **51** as illustrated in FIG. **11**. Since the portion on the side of the end portion **4d** of the display member **3** is held by the holder **50**, the display member **3** in the non-displaying position can maintain its position without moving in the gravitational direction and without causing rattling even when the vibration of the vehicle or the like acts.

[0061] As a modified example of the present disclosure, in each of the displaying-side restrictor **40a** and the non-displaying-side restrictor **40b**, the cantilever **28a** and the roller **27** may be supported by the supports **2a** and **2b**, and the displaying-side restriction surface **41** and the non-displaying-side restriction surface **42** may be supported by the display member **3** or the attached member **25**. In this modified example, in FIGS. **12A** and **12B**, the restriction surfaces **41** and **42** move in the front-rear direction (Y1-Y2 direction) without moving the roller **27**. However, also in this case, when the moving direction of the roller **27** relative to each of the restriction surfaces **41** and **42** is the restriction proceeding direction Sy1 and the restriction proceeding direction Sy2, respectively, the contact portion T between the roller **27** and the restriction surface is positioned further toward the front in each of the restriction proceeding direction Sy1 and the restriction proceeding direction Sy2, than the center line o in the free state of the cantilever **28a**, and a moving load resistance due to the flexural elastic force of the cantilever **28a** can be reduced.

[0062] Furthermore, the restrictor including the cantilever **28a**, the roller **27**, and the restriction surface may be used only by either of the displaying-side restrictor **40a** or the non-displaying-side restrictor **40b**, and another type of restrictor may be used to restrict the other of the displaying-side restrictor **40a** or the non-displaying-side restrictor **40b** of the display member **3**. It should be noted that the cantilever is not limited to a plate shape, and may have a circular or elliptical cross section. The spring for pressing the roller **27** against the restriction surface may be a torsion spring or a coil spring instead of a cantilever.

[0063] In the above embodiment, the mechanical linkage **30** is used as the display-position switcher for switching the position of the display member **3**, but instead of the mechanical linkage **30**, a rotary cam member and an arm or the like for switching the position of the display member **3** by transmitting the rotational force of the rotary cam member may be used.

[0064] In the restrictor of the movable display device of the present disclosure, the roller and the restriction surface are pressed against each other, the roller is rolled, and the display member is brought to at least one of the displaying position or the non-displaying position. Therefore, the load required to put the display member into each position becomes a rolling load, and the display member can be stabilized and restricted at a low load in the displaying position and in the non-displaying position.

[0065] In the movable display device of the present disclosure, when the slide member moves between the predetermined displaying position and the predetermined non-displaying position, the movement of the display member is not restricted either by the displaying-side restrictor or the non-displaying-side restrictor. Therefore, even in a movable structure with a relatively large load in which the display member is rotationally moved from the displaying position, in which the end of the display member faces the gravitational direction, to the non-displaying position, in which the end of the display member is rotationally lifted upward, the load by the restrictor can be prevented from acting constantly, and the operation load can be prevented from becoming excessive.

Claims

1. A movable display device, comprising: a support; a display member; and a driver configured to rotationally move the display member to be placed into a displaying position and a non-displaying position, wherein a restrictor configured to restrict a movement of the display member upon bringing the display member into at least one of the displaying position or the non-displaying position is provided, and the restrictor includes a roller provided on or with either of the support, or the display member or an attached member which rotationally moves with the display member; a restriction surface provided on or with another of the support, or the display member or the attached member; and a spring configured to press the roller against the restriction surface.
2. The movable display device according to claim 1, wherein the spring is a cantilever that is elastically deformable, and the roller is supported on a free end of the cantilever, and in conjunction with a moving direction of the roller relative to the restriction surface upon reaching at least one of the displaying position or the non-displaying position by the display member, the moving direction upon the reaching being a restriction proceeding direction, the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further ahead in the restriction proceeding direction than a fixed end of the cantilever, while the roller is pressed against the restriction surface.
3. The movable display device according to claim 2, further comprising: a displaying-side restrictor configured to restrict a movement of the display member upon reaching the displaying position by the display member; and a non-displaying-side restrictor configured to restrict the movement of the display member upon reaching the non-displaying position by the display member, wherein upon restricting the display member by the displaying-side restrictor or by the non-displaying-side restrictor, the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further ahead in the restriction proceeding direction than the fixed end of the cantilever.
4. The movable display device according to claim 3, wherein the driver includes a slide member in which the display member is rotatably supported; a power source configured to reciprocate the slide member to move the slide member between a predetermined displaying position and a predetermined non-displaying position; and a display-position switcher configured to rotationally move the display member toward the displaying position by using a moving force of the slide member upon moving the slide member toward the predetermined displaying position, and to rotationally move the display member toward the predetermined non-displaying position by using the moving force of the slide member upon moving the slide member toward the non-displaying position, wherein the displaying-side restrictor and the non-displaying-side restrictor are arranged with an interval in a moving direction of the slide member.
5. The movable display device according to claim 4, wherein the cantilever and the roller are supported by the display member or an attached member that rotationally moves with the display member, and for both cases where the display member is restricted by the displaying-side restrictor and by the non-displaying-side restrictor, by switching a direction of the cantilever upon moving of the slide member between the predetermined displaying position and the predetermined non-

displaying position, the cantilever is inclined in the restriction proceeding direction such that the free end of the cantilever is positioned further ahead in the restriction proceeding direction than the fixed end of the cantilever.

6. A movable display device, comprising: a support; a slide member supported to reciprocate in a direction crossing a gravitational direction with respect to the support; a display member rotatably supported by the slide member; a power source configured to reciprocate the slide member toward a predetermined displaying position and a predetermined non-displaying position; a display-position switcher configured to rotationally move the display member toward the displaying position, in which an end of the display member opposite to a rotation support side of the display member faces the gravitational direction, by using a moving force of the slide member upon moving the slide member toward the predetermined displaying position, and to rotationally move the display member toward the predetermined non-displaying position, in which the end of the display member is rotationally lifted upward, by using the moving force of the slide member upon moving the slide member toward the non-displaying position; a displaying-side restrictor configured to restrict a movement of the display member in the displaying position; and a non-displaying-side restrictor configured to restrict the movement of the display member in the non-displaying position, wherein upon moving the slide member between the predetermined displaying position and the predetermined non-displaying position, the movement of the display member is not restricted either by the displaying-side restrictor or the non-displaying-side restrictor.

7. The movable display device according to claim 6, further comprising: a guide cam configured to guide the display-position switcher upon moving the slide member, wherein the guide cam is provided with an inclined guide portion, which is inclined with respect to a moving direction of the slide member, and the display-position switcher is guided by the inclined guide portion upon moving the slide member between the predetermined displaying position and the predetermined non-displaying position, and upon guiding the display-position switcher by the inclined guide portion, the display member is not restricted either by the displaying-side restrictor or the non-displaying-side restrictor.

8. The movable display device according to claim 6, wherein the displaying-side restrictor and the non-displaying-side restrictor include a roller provided on or with either of the support, or the display member or an attached member which rotationally moves with the display member; a restriction surface provided on or with another of the support, or the display member or the attached member; and a spring which presses the roller against the restriction surface.

9. The movable display device according to claim 6, further comprising: a holder configured to hold the display member rotationally moved to be placed into the non-displaying position, so as not to move in the gravitational direction by using the moving force of the slide member upon moving the slide member toward the predetermined non-displaying position.
